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 NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)


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payment status

 Course
outline

 About
NPTEL ()

 How does an
NPTEL
online
course
work? ()

 Week 1
Introduction
and
Principles of
Algorithms ()

 Week 2
Shortest
Path
Algorithms
and
problems ()

 Week 3
Bellman
Equations ()

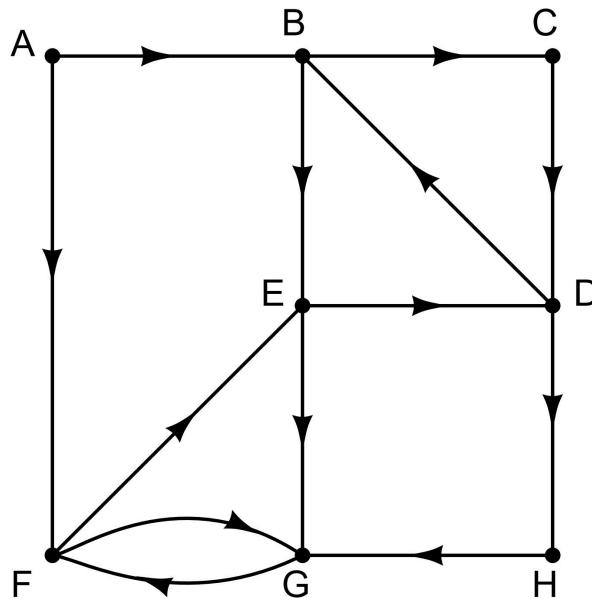
Week 7 Assignment 7

The due date for submitting this assignment has passed.

Due on 2025-09-10, 23:59 IST.

Assignment submitted on 2025-09-09, 23:37 IST

- 1) Perform DFS on the given graph. Whenever there is a choice of vertices pick **1 point** the one that is alphabetically first.



State true or false:
(A,F) is a back edge.

- ☒ False
☐ True

Yes, the answer is correct.

Week 4
Bellman
Ford and
Dijkstra
Algorithm ()

Week 5 All
Pair Shortest
Path ()

Week 6
Prims and
Kruskal's
Algorithm ()

Week 7 DFS
and Cut
Vertex ()

☐ DFS (unit?
 unit=24&lesso
 n=82)

☐ DFS (unit?
 unit=24&lesso
 n=83)

☐ Algorithm for
 Cut Vertex
 (unit?
 unit=24&lesso
 n=84)

☐ Iterative DFS
 (unit?
 unit=24&lesso
 n=85)

☐ DFS in
 Directed
 Graph (unit?
 unit=24&lesso
 n=86)

☐ Week 7
 Lecture
 Material (unit?
 unit=24&lesso
 n=68)

☒ **Quiz: Week 7**
Assignment 7
(assessment?
name=103)

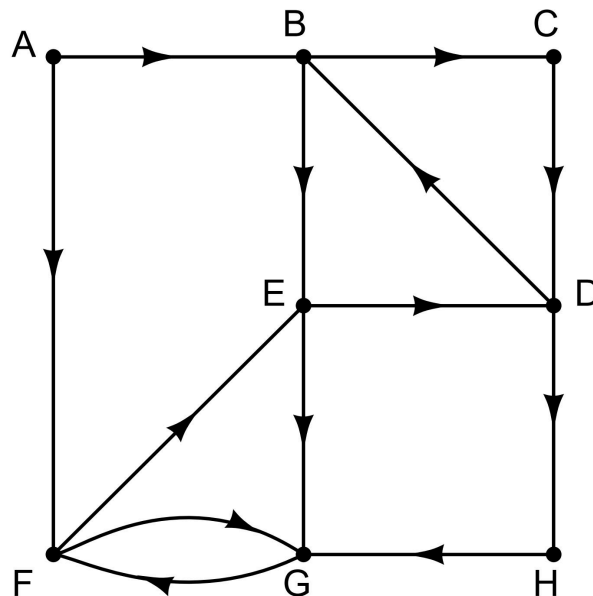
☐ Feedback
 Form (unit?)

Score: 1

Accepted Answers:

False

- 2) Perform DFS on the given graph. Whenever there is a choice of vertices pick **1 point** the one that is alphabetically first.



State true or false:

(B, E) is a forward edge.

☐ False

☒ True

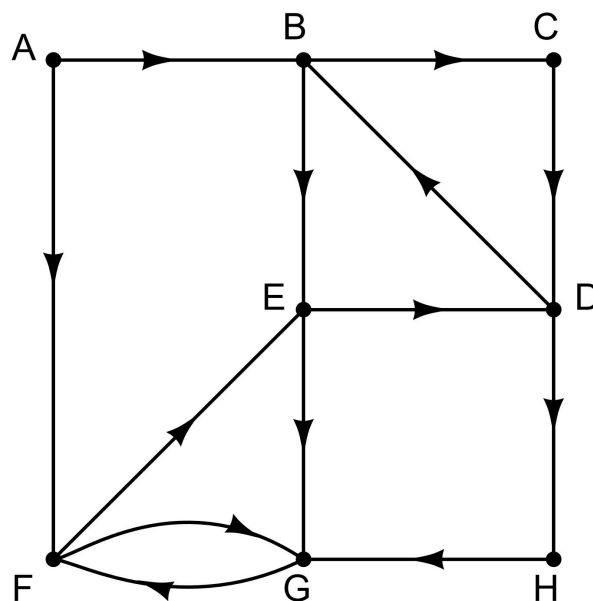
Yes, the answer is correct.

Score: 1

Accepted Answers:

True

- 3) Perform DFS on the given graph. Whenever there is a choice of vertices pick **1 point** the one that is alphabetically first.



unit=24&lesson=32)

Week 8
Strong
Connected
Components
& BFS ()

Video
download ()

Live session
()

State true or false:

(G, F) is a tree edge .

☐ False

☒ True

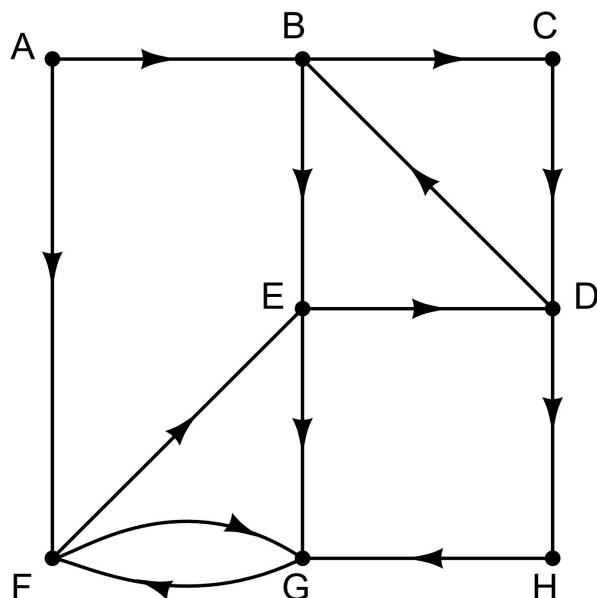
Yes, the answer is correct.

Score: 1

Accepted Answers:

True

- 4) Perform DFS on the given graph. Whenever there is a choice of vertices pick **1 point** the one that is alphabetically first.



State true or false:

(F, G) is a cross edge .

☒ False

☐ True

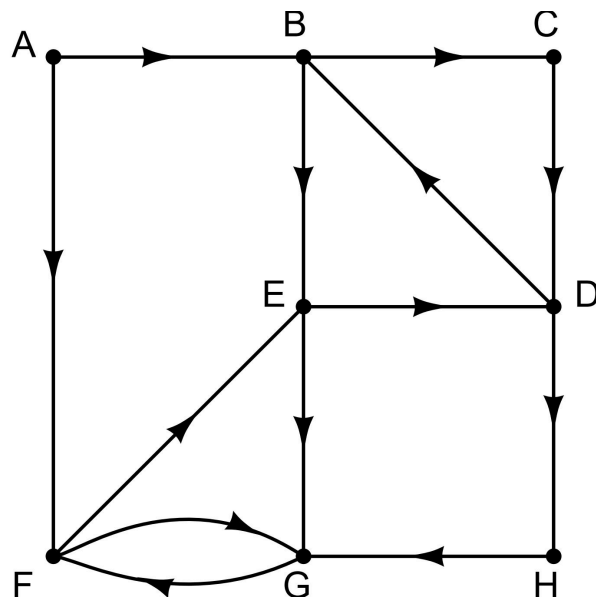
Yes, the answer is correct.

Score: 1

Accepted Answers:

False

- 5) Perform DFS on the given graph. Whenever there is a choice of vertices pick **1 point** the one that is alphabetically first.



State true or false:

Number of back edges are 5.

☒ False

☐ True

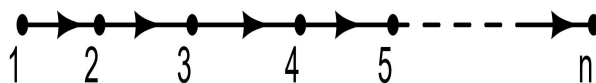
Yes, the answer is correct.

Score: 1

Accepted Answers:

False

6) Consider DFS on the given directed graph.



Perform DFS starting from n in the decreasing order of vertex label.

Number of backedges we obtain in the DFS is ____

0

Yes, the answer is correct.

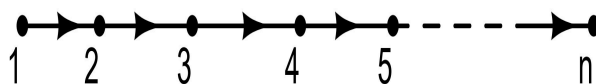
Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

7) Consider DFS on the given directed graph.



Perform DFS starting from n in the decreasing order of vertex label.

Number of forward edges we obtain in the DFS is ____

0

Yes, the answer is correct.

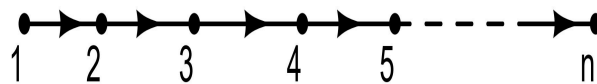
Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

8) Consider DFS on the given directed graph.



Perform DFS starting from n in the decreasing order of vertex label.

Number of tree edges we obtain in the DFS is ____

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

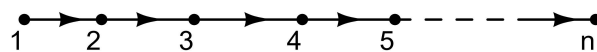
1 point



1 point



9) Consider DFS on the given directed graph.



Perform DFS starting from n in the decreasing order of vertex label.

Number of forests we obtain in the DFS is ____

☒ n
☐ $n-1$
☐ $n+1$
☐ 1

Yes, the answer is correct.

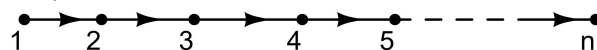
Score: 1

Accepted Answers:

n

1 point

10) Consider DFS on the given directed graph.



Perform DFS starting from n in the decreasing order of vertex label.

Number of crossed edges we obtain in the DFS is ____

☐ n
☒ $n-1$
☐ $n+1$
☐ 1

Yes, the answer is correct.

Score: 1

Accepted Answers:

$n-1$